

Weekly Progress Report

AgAI Pesticide Residue & Leaf Classification Project

Week 2: 3/9/2026

Progress This Week

1. Refined baseline modeling objective

Following Dr. Chen's suggestion, I redesigned the initial baseline from a **binary pesticide decision model** to a **multi-class classification model predicting pesticide concentration bins** (exact bins to be finalized with the team).

The updated baseline objective is:

- Model: CNN $\rightarrow$ MLP $\rightarrow$ multiclass output
- Loss: CrossEntropyLoss
- Goal: predict which pesticide concentration bin a leaf image belongs to.

This change allows the model to capture more granular pesticide level differences rather than a simple threshold decision.

2. Began implementing baseline training pipeline

I started building the full training pipeline, including:

- Image preprocessing and dataloader
- Image resizing and normalization
- Label mapping system for configurable pesticide bins
- Small CNN baseline architecture (currently 4 convolutional blocks)
- Final linear layer with configurable number of classes
- Training loop using CrossEntropyLoss
- Evaluation metrics including confusion matrix and per-class accuracy

This baseline CNN is intentionally shallow to determine whether the dataset contains usable predictive signal before exploring more complex architectures or pretrained backbones.

3. Reviewed William's previous pipeline and detection model

I received William's repository and model weights from last year's project and began reviewing the architecture and preprocessing approach.

One important component is his **RF-DETR model fine-tuned for leaf detection**.

Because the RF-DETR detector was trained on visually similar leaf images, it is a reasonable starting point for isolating the primary leaf in each image before classification.

The planned evaluation process is:

Phase 1:

- Run RF-DETR using William's pretrained weights
- Evaluate qualitatively on a subset of the pesticide dataset

Key evaluation criteria:

- Does it consistently detect the primary leaf?
- Does it correctly select the main leaf when multiple leaves are present?
- Are bounding boxes tight enough for cropping?
- Does performance degrade on leaves with visible pesticide residue?

If performance is satisfactory, the pipeline will proceed with:

leaf detection → crop → resize → classification

If detection performance is insufficient, I will fine-tune a task-specific detector (likely YOLO) on our dataset.

4. Dataset inspection and preliminary preprocessing

While waiting for access to the Roboflow annotations, I located last year's pesticide dataset (~580 images across 400, 600, 800, and 1000 ppm classes).

I began implementing preprocessing and pipeline components using this dataset as a reference for:

- image dimensions
- preprocessing requirements
- CNN architecture design

This allows development of the training pipeline while waiting for the current year's annotated dataset.

Observations / Potential Dataset Concerns

During dataset inspection, it appears possible (based on the folder structure of last year's dataset) that **each pesticide concentration may correspond to a single plant.**

If the structure is:

- Plant A → 400 ppm
- Plant B → 600 ppm
- Plant C → 800 ppm
- Plant D → 1000 ppm

then the model may risk learning **plant identity rather than pesticide level**, because images from the same plant share:

- leaf texture
- lighting conditions
- background context
- biological characteristics

In that scenario, classification performance could be artificially inflated if leaves from the same plant appear in both training and test sets.

This may limit the reliability of classification-based evaluation and may require alternative approaches such as:

- residue segmentation
- residue coverage estimation
- feature-based analysis

I plan to discuss this with the team during the upcoming meeting with Melissa.

Plan for Next Week

1. Run and evaluate William's RF-DETR leaf detector

- Test qualitative performance on pesticide images
- Determine whether fine-tuning is necessary.

2. Complete baseline CNN pipeline

- finalize preprocessing pipeline
- complete training and evaluation loops
- run initial baseline experiments.

3. Confirm dataset structure and annotations

- verify whether pesticide labels are plant-level or leaf-level
- inspect Roboflow annotations once access is granted.

4. Discuss dataset structure and experimental design with the chemistry team

- clarify number of plants per pesticide level
- determine whether additional data collection may be needed.